DETAILED CASE STUDY · IRON MAKING

# SAIL DURGAPUR STEEL PLANT

A study of how SAIL's Durgapur plant makes iron: where its ore and coal come from, how its coke ovens and sinter plant prepare the burden, how its three British-designed blast furnaces work, and what its planned new furnace will change.

Data up to October 2026 · Mt = million tonnes · thm = tonne of hot metal · BF = blast furnace

01

## EXECUTIVE SUMMARY

**Durgapur Steel Plant (DSP)** is one of the five integrated steel plants of the **Steel Authority of India Ltd (SAIL)**. It stands on the banks of the Damodar river in Paschim Bardhaman district, West Bengal, about 158 km from Kolkata. It was set up in the late 1950s with help from the **United Kingdom** and started production in 1959. In the same years India built Bhilai with Soviet help and Rourkela with German help.

This study focuses on the **iron zone**: everything up to the liquid iron ("hot metal") that leaves the blast furnace. Durgapur makes its own coke in five oven batteries and its own sinter on three machines, then smelts them in **three blast furnaces** (BF-2, BF-3 and BF-4) with a rated capacity of **2.40 million tonnes of hot metal a year**. In 2022 to 2024 it produced more than that rating. In July 2026 it had its best July ever, at 2,27,400 tonnes of hot metal.

The central story is that a plant with **small, old furnaces** (built 1959 to 1967) keeps them productive through good burden preparation and relining, and now plans a single large new furnace, **BF-5**, that will make nearly as much iron as all three existing ones.

| 2.40 Mt | 3 | 4,600 $m^3$ | 5 |
|---|---|---|---|
| | | Total useful volume | |

Hot metal capacity a year

Blast furnaces in operation

Coke oven batteries, 78 ovens each

**3.30 Mt**
Sinter capacity a year

**~2.2 Mt**
Crude steel capacity a year

02
# HISTORY & MILESTONES

### Late 1950s
India's second Five-Year Plan sets up three public steel plants with foreign help: Bhilai (USSR), Rourkela (West Germany) and Durgapur (United Kingdom). Durgapur's blast furnaces are supplied by the British firm **Head Wrightson**.

### 1959
Production begins with a capacity of 1 Mt of crude steel a year. BF-2 dates from this first phase.

### 24 April 1960
The first steel ingot is cast in the original steel melting shop.

### 1967
BF-4, the plant's largest furnace (1,800 m³ useful volume), comes into operation.

### 1967–69
Capacity is raised to 1.6 Mt of crude steel a year.

### 1990s
A modernisation programme announced in 1985 is carried out, adding new facilities and raising capacity to 1.8 Mt. BF-2 is relined in December 1993 and BF-4 in April 1996.

### 2000s–2010s
The Modernisation & Expansion Plan (MEP) takes crude steel capacity to about 2.2 Mt and adds a 1 Mt medium structural mill.

### 2024
Brownfield expansion is briefly put on hold while SAIL reviews spending, then confirmed as approved in October 2024.

### March 2025
BF-3 completes its latest relining.

### 2025
Stage-I approval granted; tendering under way. The environmental report outlines BF-5 and a pellet plant.

### ~2030
Planned start of BF-5 (about 3,000 m³, 2.36 Mt a year) and a 2.5 Mt pellet plant.

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03

# THE IRONMAKING ROUTE AT DURGAPUR

Durgapur follows the classic **blast furnace (BF) – basic oxygen furnace (BOF)** route. The iron zone has four main units, each feeding the next.

**01** **Raw Materials Handling Plant**
Receives ore, flux and coal by rail; prepares a uniform blend for the sinter plant and supplies lump materials to the furnaces and steel shop.

**02** **Coke Ovens & Coal Chemicals**
Converts coking coal into hard, porous coke and recovers by-products from the gas.

**03** **Sinter Plant**
Converts ore fines, coke breeze and limestone into sinter, the main iron-bearing feed for the furnaces.

**04** **Blast Furnaces**
Smelt sinter, lump ore and coke into liquid hot metal and slag. Hot metal goes to the steel melting shop.

### MATERIAL FLOW IN ONE YEAR

| STAGE | TONNES A YEAR | SOURCE |
|---|---|---|
| Iron ore lump and fines in | ~2.9 Mt | Older published estimate |
| Coal in | ~1.84 Mt | Older published estimate |
| Total raw materials in | ~7.4 Mt | Older published estimate |
| Coke made (capacity) | 1.7 Mt | Global Energy Monitor |
| Sinter made (capacity) | 3.30 Mt | SAIL |
| Hot metal made (2024) | 2.50 Mt | Global Energy Monitor |
| Crude steel made (2024) | 2.26 Mt | Global Energy Monitor |

The raw-material figures come from an older description of the plant and may not match today's exact use. They are shown for scale.

04

# RAW MATERIALS

| MATERIAL | SOURCE | ROLE IN IRONMAKING |
|---|---|---|
| **Iron ore** (hematite) | SAIL captive mines: mainly **Bolani** (Odisha), with Gua, Kiriburu and Meghahatuburu (Jharkhand) and Barsua, Taldih and Kalta (Odisha) | Source of iron, used as lump ore and, after sintering, as fines |
| **Coking coal** | Jharia-Raniganj belt (BCCL Chasnala, CCL), plus imported low-ash prime coking coal | Made into coke: fuel, reducing agent and burden support |
| **Limestone, dolomite** | Purchased flux | Combine with silica and alumina to form a fluid slag |
| **Others** | Manganese ore, bauxite, ferro-alloys | Mostly for steelmaking and chemistry control |

### LOCATION ADVANTAGE

Durgapur sits on the edge of India's main coking-coal region, the **Jharia-Raniganj belt**. Coal is the bulkiest input after ore, so being near it cuts freight. Ore comes from SAIL's own mines in Odisha and Jharkhand, so the plant doesn't depend on buying ore on the open market.

### THE COAL PROBLEM

Indian coking coal has a lot of ash, which weakens coke and adds to slag volume. So Durgapur, like other Indian integrated plants, blends in **imported low-ash coking coal**. This makes coal prices one of the biggest drivers of ironmaking cost.

Ore quality is managed at the source: lump ore and fines are **beneficiated (washed) at Bolani**, and lump ore is screened again at the plant.

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05

## RAW MATERIALS HANDLING

The Raw Materials Handling Plant (RMHP) is the starting point. Consistent chemistry and size of raw material are what let a blast furnace run steadily.

- **3 wagon tipplers** unload ore, flux and coal from rail wagons.
- **3 blender-cum-reclaimers** lay materials in long piles and reclaim them across the pile, averaging out variations in chemistry. This "bedding and blending" gives the sinter plant a uniform mix.
- Lump ore and flux are sent directly to the furnaces and steel shop.
- Coal is **selectively crushed** at the coal handling plant, and silo-cum-blending bunkers prepare the coal blend for the ovens.

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06

## COKE MAKING

Coke does three jobs in a blast furnace: it is the **fuel** (burning at the tuyeres for heat), the **reducing agent** (its carbon makes CO gas), and the **support** (strong lumps keep the burden open so gas can flow, even where everything else has melted). No other material can do the third job.

| 5 | 78 | 4.45 m | 1.7 Mt |
|---|---|---|---|
| Batteries: 4 in operation, 1 under cold repair | Ovens in each battery | Oven height | Coke capacity a year |

**01** **Coal preparation**

Indian and imported coals are crushed and blended for the right strength and ash content.

**02** **Carbonisation**

The blend is heated in narrow ovens without air, typically to about 1,000 to 1,100°C for many hours. Volatile matter leaves as gas; hard, porous coke remains.

**03** **Coke handling**

Coke is pushed out, cooled and screened.
**Blast-furnace-grade coke**
goes to the furnaces;
**undersized coke**
(breeze) goes to the sinter plant as fuel.

**04** **Coal chemicals**

The gas is cleaned and by-products are recovered: naphthalene oil, heavy creosote oil, light oil, crude tar, nitration-grade benzene and toluene, solvent naphtha, and "Raja" brand fertiliser.

The cleaned **coke oven gas** is mixed with blast furnace gas and BOF gas and piped around the plant as fuel. It also supplies the neighbouring **Alloy Steels Plant**.

Temperatures are typical industry values; SAIL does not publish Durgapur's.

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07

# SINTERING

Mining and crushing produce many ore **fines** too small to charge into a blast furnace, because they would block the gas flow. Sintering turns them into strong, porous lumps. SAIL notes that **a high percentage of sinter in the charge is a prerequisite** for good

blast furnace productivity at Durgapur.

| MACHINE | HEARTH AREA |
|---|---|
| Sinter machine 1 | 142.7 m² |
| Sinter machine 2 | 142.7 m² |
| Sinter machine 3 | 198 m² |
| **Total capacity** | **3.30 Mt a year** |

**01** **Blend mix**

Ore fines, limestone, coke breeze and recycled fines from the raw material yard.

**02** **Mixing and balling**

Water is added in drums so the mix forms small balls that let air pass through.

**03** **Ignition and sintering**

The mix is spread on a moving strand of pallets and ignited. Fans pull air down through the bed; the flame front moves down, reaching about 1,300°C, and bonds the particles.

**04** **Crushing, cooling, screening**

The sinter cake is broken and cooled; undersize returns to the mix.

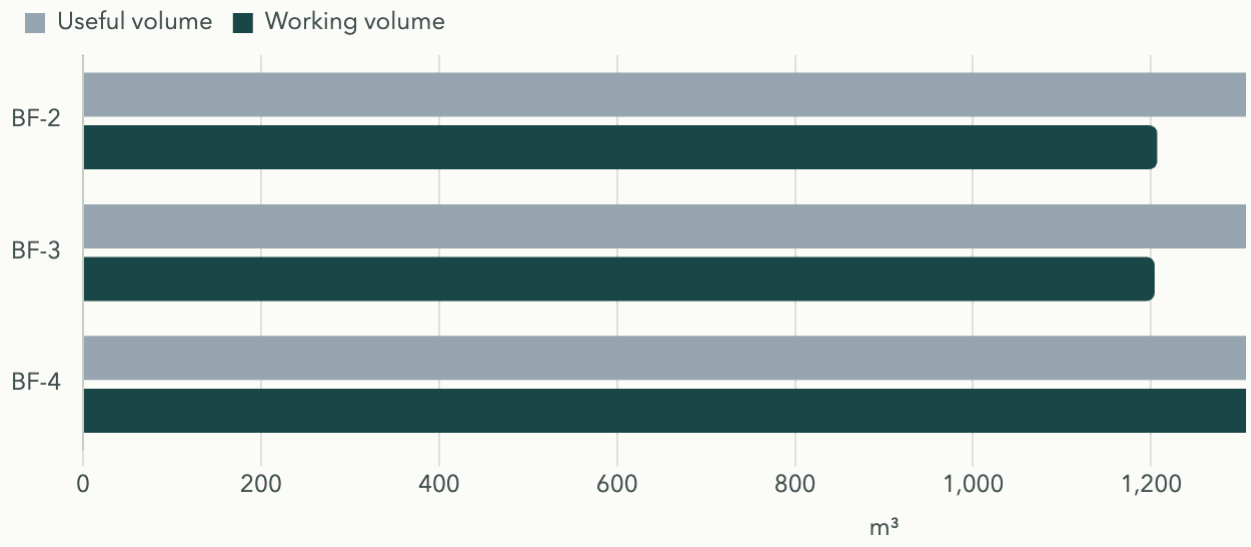
**Why sinter is better than raw ore:** it is already partly fluxed, so less raw limestone goes into the furnace; it is porous, so gas reduces it quickly; and it uses fines that would otherwise be waste.

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# DURGAPUR'S BLAST FURNACES

| FURNACE | USEFUL VOLUME | WORKING VOLUME | PRODUCTIVITY* | STARTED | LAST RELINED |
|---|---|---|---|---|---|
| **BF-2** | 1,400 m³ | 1,207 m³ | 1.71 | 1959 | Dec 1993 |
| **BF-3** | 1,400 m³ | 1,204 m³ | 1.82 | Not published | Mar 2025 |
| **BF-4** | 1,800 m³ | 1,539 m³ | 1.69 | 1967 | Apr 1996 |

*Tonnes of hot metal per m³ of working volume per day, as published by SAIL. Volumes and productivity from SAIL; start and reline dates from Global Energy Monitor. BF-3 has a bell-less top. BF-1, from the original plant, is no longer in service. Global Energy Monitor lists BF-2 as "operating pre-retirement".




Useful and working volume of each furnace, m³.

## USEFUL VS WORKING VOLUME

**Useful volume** is the inside volume from the taphole level up to the stockline (the top of the burden). **Working volume** is smaller: it leaves out the hearth below the tuyeres, where liquid iron and slag collect, so it measures where the burden actually reacts. SAIL quotes Durgapur's productivity against working volume. Always check which volume is used when comparing furnaces.

## RELINING

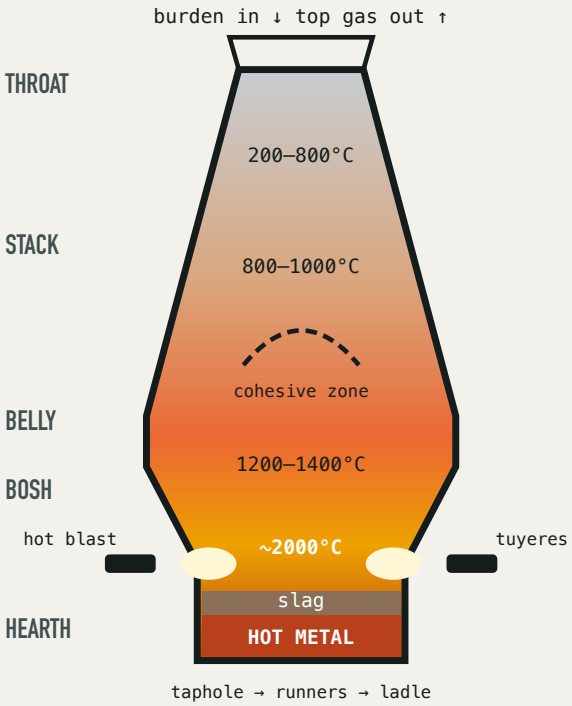
A **relining** (or capital repair) is the major shutdown where a furnace's worn refractory lining is replaced and equipment is upgraded. BF-3's relining in March 2025 is recent, while BF-2 and BF-4 were last relined in the 1990s. This helps explain why BF-3 has the highest productivity and why BF-2 is marked for retirement.

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09

# BLAST FURNACE BASICS

A blast furnace is a tall shaft furnace that works on the **counter-current principle**: solid burden is charged at the top and moves down, while hot reducing gas made at the bottom rises through it. As they pass, the gas heats the burden and removes oxygen from the iron oxide. Liquid iron and slag collect in the hearth and are tapped regularly.




## PARTS OF THE FURNACE

- **Throat:** the top, where burden is charged in alternating layers of coke and iron-bearing material.
- **Stack:** widens downward so the burden can expand as it heats and doesn't jam. Most gas reduction happens here.
- **Belly:** the widest part.
- **Bosh:** narrows downward because the burden softens, melts and shrinks. The cohesive zone, where ore softens, sits here.
- **Tuyeres:** water-cooled copper nozzles that blow in hot air from the stoves.
- **Hearth:** the bottom, where liquid iron sits under a layer of lighter slag.

Temperatures are typical textbook ranges.

## HOT BLAST STOVES AND GAS CLEANING

Air is heated in tall regenerative **stoves** lined with checker bricks, which are first heated by burning blast furnace gas, then used to heat the air, typically to about 1,000 to 1,200°C. The **top gas** leaving the furnace is dusty but still contains CO, so it is cleaned

and used as fuel.

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10

# CHEMISTRY OF IRONMAKING

## 1 · COMBUSTION AT THE TUYERES

| | |
|---|---|
| $C + O_2 \rightarrow CO_2$ | strongly exothermic |
| $CO_2 + C \rightarrow 2CO$ | Boudouard / solution loss, endothermic |
| Net: $2C + O_2 \rightarrow 2CO$ | flame temperature ~2,000°C+ |

## 2 · INDIRECT REDUCTION (BY CO GAS, IN THE STACK)

| | |
|---|---|
| $3Fe_2O_3 + CO \rightarrow 2Fe_3O_4 + CO_2$ | hematite → magnetite |
| $Fe_3O_4 + CO \rightarrow 3FeO + CO_2$ | magnetite → wüstite |
| $FeO + CO \rightarrow Fe + CO_2$ | wüstite → iron |

## 3 · DIRECT REDUCTION (BY CARBON, LOWER DOWN)

| | |
|---|---|
| $FeO + C \rightarrow Fe + CO$ | above ~1,000°C; strongly endothermic |

Direct reduction takes more heat, and therefore more coke, than indirect reduction. Good practice aims to do as much reduction as possible indirectly in the stack, which is why **porous sinter and even gas flow** matter so much.

## 4 · FLUX AND SLAG

| | |
|---|---|
| $CaCO_3 \rightarrow CaO + CO_2$ | calcination |
| $CaO + SiO_2 \rightarrow CaSiO_3$ | slag; $Al_2O_3$ and MgO also join |

## 5 · OTHER ELEMENTS

- **Silicon:** $SiO_2 + 2C \rightarrow Si + 2CO$ at high hearth temperatures; hot metal silicon shows how hot the hearth is.
- **Sulphur**, mainly from coke, is taken up by basic slag as CaS.

- **Carbon pick-up:** iron absorbs carbon as it trickles through the coke, lowering its melting point to about 1,150°C and ending at roughly 4 to 4.5% C.

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11

# HOT METAL & SLAG

### HOT METAL (TYPICAL)

| ELEMENT | RANGE |
|---|---|
| Carbon | 4.0–4.5% |
| Silicon | 0.4–0.8% |
| Manganese | 0.1–0.3% |
| Phosphorus | 0.1–0.15% |
| Sulphur | 0.03–0.05% |
| Temperature | 1,450–1,500°C |

### BF SLAG (TYPICAL)

| OXIDE | RANGE |
|---|---|
| CaO | 30–35% |
| $SiO_2$ | 32–35% |
| $Al_2O_3$ (high in India) | 17–22% |
| MgO | 8–10% |
| Basicity $CaO/SiO_2$ | ~1.0–1.1 |

Typical values for Indian blast furnaces from standard ironmaking texts. Durgapur's exact chemistry is not published.

At Durgapur, the **blast furnace slag is granulated** (quenched with water into sand-like grains) and sold to cement companies, where it replaces part of the clinker.

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12

# FROM HOT METAL TO PRODUCTS

Although this study is about iron, it helps to know where the hot metal goes.

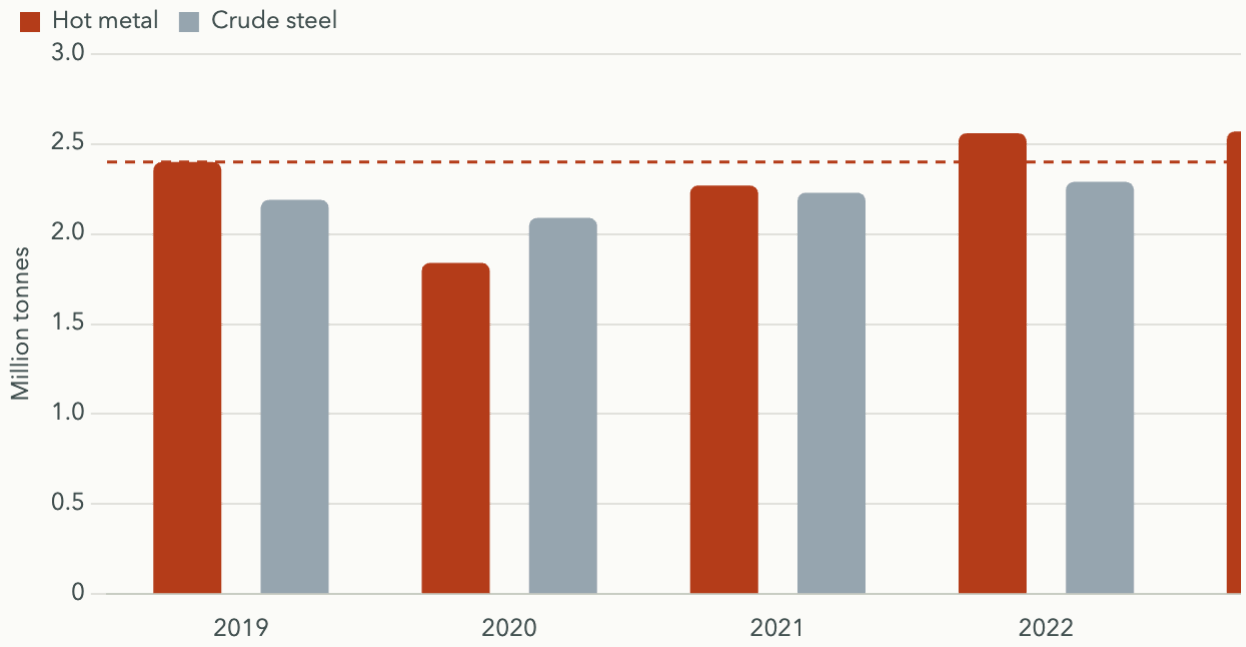
- **Steel Melting Shop:** 3 basic oxygen furnaces (110/130 t), 1 vacuum arc degassing unit and 3 ladle furnaces, with a nominal heat size of 120 t. Oxygen blown into the converter burns off excess carbon and silicon to make steel.
- **Continuous casting:** 2 billet casters, 1 bloom caster and 1 bloom-cum-round caster. Vacuum-degassed steel is bottom-poured into ingots for loco wheels and for the Alloy Steels Plant.

- **Wheel & Axle Plant:** forged and machined wheels and axles for Indian Railways locomotives, coaches, LHB and metro coaches.
- **Rolling mills:** earthquake-resistant TMT bars (merchant mill, 3.6 lakh t a year), a new 1 Mt medium structural mill for beams, channels and angles, and a section mill.

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13

# PRODUCTION PERFORMANCE




Hot metal and crude steel production, million tonnes (Global Energy Monitor). The dashed line is the rated hot metal capacity of 2.40 Mt.

- **Above rated capacity:** hot metal reached 2.56 Mt in 2022 and 2.57 Mt in 2023, about 7% above the 2.40 Mt rating.
- **COVID dip:** output fell to 1.84 Mt in 2020 and recovered the next year.
- **July 2026 record:** 2,27,400 t of hot metal (105% of the annual business plan for the month) and 2,01,200 t of crude steel, the best July ever. Coke oven pushing also set a best July.

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# WORKED CALCULATIONS

These use Durgapur's published numbers.

### 1 · DAILY OUTPUT FROM PRODUCTIVITY

Daily output = productivity × working volume.

BF-2: 1.71 × 1,207 = 2,064 t/day
BF-3: 1.82 × 1,204 = 2,191 t/day
BF-4: 1.69 × 1,539 = 2,601 t/day

Total ≈ **6,856 t/day**

### 2 · CHECK AGAINST YEARLY CAPACITY

6,856 × 365 = 2,502,000 t ≈ **2.50 Mt a year**, close to the 2.40 Mt rated capacity and the 2.50 Mt actually made in 2024.

### 3 · JULY 2026 PERFORMANCE

2,27,400 t ÷ 31 days = **≈ 7,335 t/day**, about 7% above the daily figure from published productivity.

### 4 · PRODUCTIVITY ON USEFUL VOLUME

BF-3: 2,191 ÷ 1,400 = **1.56 t/m³/day**, lower than 1.82 because useful volume is bigger than working volume. That's why you must state which volume you use.

### 5 · SINTER PER TONNE OF IRON

3.30 Mt sinter ÷ 2.40 Mt hot metal = **≈ 1.38 t sinter per thm**, showing that sinter is the main iron-bearing feed.

### 6 · SMALL VS LARGE FURNACE

Bhilai's BF-8 (4,060 m³) is rated at 8,030 t/day. 8,030 ÷ 6,856 = **1.17×** all three Durgapur furnaces combined.

### 7 · WHAT BF-5 ADDS

After expansion: 3.42 Mt iron capacity ÷ 2.40 Mt today = **+43%** more iron, even though BF-2 retires and BF-4 goes on standby.

Calculations are our own, using the figures in this study.

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15

# ENVIRONMENT & BY-PRODUCTS

- **Slag to cement:** granulated blast furnace slag is sold to cement makers, saving limestone and $CO_2$ in cement production.
- **Gas reuse:** blast furnace, coke oven and BOF gases are mixed and used as fuel across DSP and the Alloy Steels Plant.
- **Coal chemicals:** tar, oils, benzene, toluene, solvent naphtha and fertiliser are recovered instead of being released.
- **Certifications:** ISO 14001 (environmental management) and ISO 50001 (energy management).
- **Pellet plant (planned):** will let the plant use fine ore that sintering can't easily handle.
- **Carbon challenge:** the blast furnace route is coal-based and emits a lot of $CO_2$. Bigger, more efficient furnaces and better burden are near-term ways to cut emissions per tonne.

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16

# SWOT · IRONMAKING

### STRENGTHS

- Captive iron ore from SAIL mines
- Close to the Jharia-Raniganj coal belt
- Full in-house coke and sinter chain
- Produced above rated capacity in 2022–24
- 65+ years of furnace operating experience

### WEAKNESSES

- Small furnaces (1,400–1,800 m³) from 1959–67
- BF-2 and BF-4 last relined in the 1990s
- One coke battery under cold repair
- Reliance on imported low-ash coal

### OPPORTUNITIES

- BF-5 with 2.36 Mt a year capacity
- 2.5 Mt pellet plant using fine ore
- Higher sinter share and better burden
- Demand for rail wheels, structurals and TMT

### THREATS

- Volatile imported coal prices
- Further delays to expansion
- Tighter carbon rules on coal-based steel
- Competition from newer private plants

17

# CHALLENGES

- **Ageing furnaces:** the furnaces date from 1959 to 1967, and BF-2 is marked for retirement. Older furnaces need more maintenance and risk unplanned stops.
- **Small size:** smaller furnaces lose more heat per tonne and need more coke and labour per tonne than large modern ones.
- **Coal:** high-ash Indian coal must be blended with imported coal, exposing costs to world prices.

- **Expansion timing:** brownfield expansion was put on hold in 2024 while SAIL reviewed spending, before being approved; tenders are now in progress.

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18

# FUTURE PLANS

According to the 2025 environmental impact report summarised by Global Energy Monitor, the expansion includes:

| ITEM | PLAN |
|---|---|
| New blast furnace BF-5 | About 3,000 m³, 2.36 Mt a year |
| Pellet plant | 2.5 Mt a year |
| Existing furnaces | BF-2 decommissioned; BF-4 on standby |
| Ironmaking capacity after | 3.42 Mt a year |
| Steelmaking capacity after | 3.09 Mt a year; steel melting shops upgraded |
| Target | Around 2030, as part of SAIL's wider expansion plan |

Earlier announcements quoted different totals (for example, a 4.73 Mt addition). The EIA figures above are the most specific published plan.

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19

# CONCLUSION & LEARNINGS

### 1 · LOCATION DECIDES COST

Durgapur was built beside the Jharia-Raniganj coal belt, with ore from SAIL's own mines. Six decades on, that remains a key advantage.

### 2 · BURDEN PREPARATION DRIVES OUTPUT

3.3 Mt of sinter and 1.7 Mt of coke a year let three small furnaces make about 2.5 Mt of iron, above their rating.

### 3 · SIZE LIMITS PRODUCTIVITY

One modern 4,000 m³ furnace out-produces all three of Durgapur's. That's why the plan is one big BF-5, not more small ones.

### 4 · RELINING KEEPS OLD FURNACES ALIVE

BF-3's 2025 relining gave it the plant's best productivity, while BF-2 is nearing the end of its life.

20

# GLOSSARY

| | |
|---|---|
| **Hot metal** | Liquid iron tapped from the blast furnace, about 4 to 4.5% carbon. Called pig iron once cast and solidified. |
| **Burden** | All solid material charged into the furnace: sinter, ore, coke and flux. |
| **thm** | Tonne of hot metal, used for rates such as kg of coke per thm. |
| **Useful volume** | Furnace volume from the taphole level to the stockline. |
| **Working volume** | Smaller volume where the burden reacts; excludes the hearth below the tuyeres. |
| **Productivity** | Tonnes of hot metal per m³ of furnace volume per day. |
| **Sinter** | Porous lumps made by fusing ore fines, flux and coke breeze. |
| **Coke breeze** | Small coke particles, used as fuel in sintering. |
| **Tuyere** | Water-cooled nozzle that blows hot air into the furnace. |
| **Bell-less top** | Charging system with a rotating chute that places burden precisely. |
| **Relining** | Major repair to replace a furnace's worn refractory lining. |
| **BOF** | Basic oxygen furnace: converter that blows oxygen into hot metal to make steel. |
| **Beneficiation** | Washing and treating ore to raise its iron content and remove waste. |

# SOURCES


1. SAIL: Durgapur Steel Plant facilities (raw materials handling, coke ovens, sinter, blast furnaces, steel shop, mills)
2. Global Energy Monitor: Durgapur plant tracker (history, furnace dates and relines, production, ore and coal sources, expansion EIA 2025)
3. Suchnaji: Durgapur's July 2026 production records (in Hindi)
4. Bharat Online: Durgapur Steel Plant history and raw materials (older raw-material figures)
5. SAIL: About Durgapur Steel Plant


Sections on reactions, furnace zones, hot metal and slag chemistry, and typical coke-making and sintering temperatures use standard ironmaking textbook values and are labelled as typical. Bhilai BF-8 figures are from SAIL and PIB. Calculations are our own.

Prepared as an academic case study for an Iron Making course.

**← Back to poster**